Cut it, branch it, solve it.

A hands-on Branch-and-Cut workshop: from the simplex method to the Traveling Salesperson Problem

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Angélica Salazar[1] Vincent Boyer[2]

Online workshop · 2 hours

[1]Universidad Autónoma de Nuevo León (UANL) [2]Tecnológico de Monterrey

How this workshop works

You will need

- A Google account → **Google Colab**
- The **Miro board**, the **GeoGebra lab** and the **notebooks** (links on the next slide and in chat)
- Pen & paper

Two problems, all along

- **E** (*explanation*): we solve it on the slides and live
- **P** (*practice*): **you** solve it with the same method
- **Orange boxes** = individual work, posted on **Miro**
- The notebooks do the arithmetic: **you focus on the concepts**

Agenda (120 min)

0:00 Welcome & check-in
0:05 **1.** Where B&C fits; IP & relaxation
0:20 **2.** The simplex method in matrix form
0:40 **3.** Branch-and-Bound
0:55 **4.** Gomory cuts
1:12 *Break (5 min)*
1:17 **5.** Chvátal–Gomory cuts
1:27 **6.** Knapsack & cover cuts
1:35 **7.** TSP case study
1:55 Wrap-up & exit ticket

Questions anytime: chat or the Miro *Parking Lot*.

Check-in (Miro, 3 min)

Your turn — Tell us where you are — 3 min — Miro frame **0 – Check-in**

1. Answer the poll: *"I can solve a small LP with the simplex method"*
   (Not yet — With notes — In my sleep)
2. One word: what do you expect from *Branch-and-Cut*? (drag a sticky from the stack)
3. Open notebook `01_Simplex_Matrix_Form.ipynb` ↗ in Colab (click it below) and run the first cell.

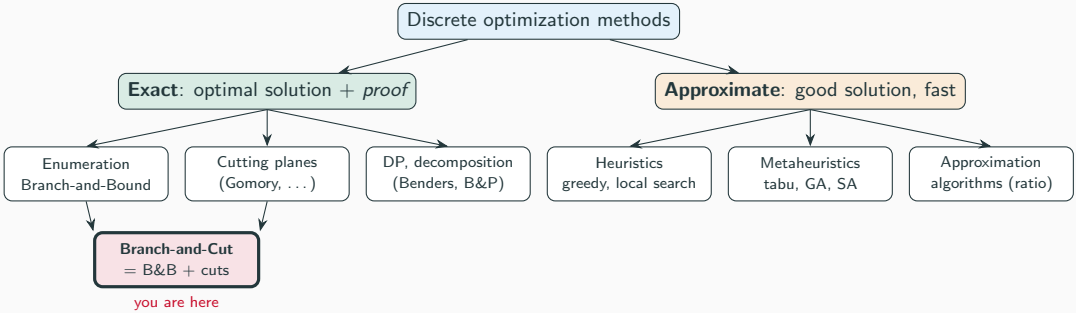
Board: `https://miro.com/app/board/uXjVEXUET_M=/` GeoGebra: GeoGebra lab ↗

Notebooks on GitHub:
`github.com/VincentALBoyer/Course-Materials/tree/main/Workshops/Branch-and-Cut`

Open in Colab: `01 Simplex` ↗ · `02 B&B` ↗ · `03 Gomory` ↗ · `04 Knapsack` ↗ · `05 TSP` ↗

1. Where Branch-and-Cut fits

Exact vs. approximate methods



Exact methods give two numbers: a **feasible solution** (lower bound, max) and a **bound** on the optimum. Their difference is the **gap**; gap $= 0$ proves optimality. Branch-and-Cut is the engine of CPLEX, Gurobi, HiGHS, SCIP, CBC.

Integer program and its LP relaxation

Integer program (IP) and relaxation (LP)

$$z_{IP} = \max\{c^\top x : Ax \le b,\ x \in \mathbb{Z}^n_{\ge 0}\}$$
$$z_{LP} = \max\{c^\top x : Ax \le b,\ x \in \mathbb{R}^n_{\ge 0}\}$$

- Dropping integrality enlarges the feasible set: $z_{LP} \ge z_{IP}$
  (a **bound**, for free).
- LP solution integer $\Rightarrow$ it is optimal for the IP.
- Otherwise: **branch** (Part 3) or **cut** (Part 4).
- Rounding the LP solution may be infeasible or far from optimal.

E — explanation problem

$$\max\ 5x_1 + 8x_2$$
$$\text{s.t. } x_1 + x_2 \le 6, \quad 5x_1 + 9x_2 \le 45$$
$$x_1, x_2 \in \mathbb{Z}_{\ge 0}$$

P — practice problem

$$\max\ 3x_1 + 4x_2$$
$$\text{s.t. } x_1 + 2x_2 \le 8, \quad 2x_1 + x_2 \le 9$$
$$x_1, x_2 \in \mathbb{Z}_{\ge 0}$$

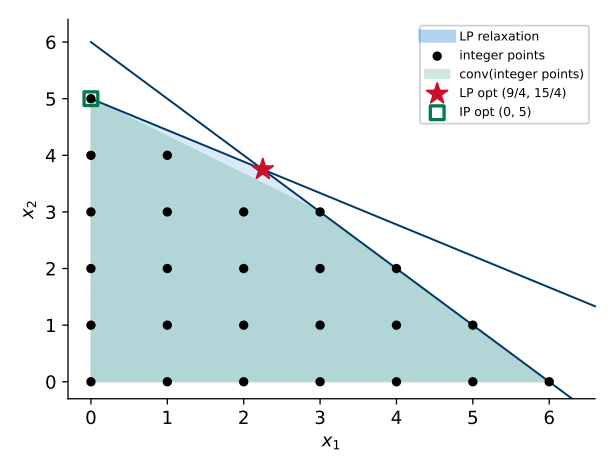
The graphical method on E

E: max $5x_1 + 8x_2$ **s.t.** $x_1 + x_2 \le 6,\ 5x_1 + 9x_2 \le 45,\ x \in \mathbb{Z}^2_{\ge 0}$

1. Draw each constraint line; keep the side that satisfies it.
2. Mark the integer points inside.
3. Slide the objective line $5x_1 + 8x_2 = k$ in the direction ∇z: the last point it touches is optimal.

- LP: $x^* = (\frac{9}{4}, \frac{15}{4})$, $z^* = 41.25$
- Round: $(2, 4)$ infeasible; $(2, 3)$ gives 34
- IP optimum: $(0, 5)$, $z = 40$, far from x^*

Live in GeoGebra: `x+y<=6`, `5x+9y<=45`, slider `k`, line `5x+8y=k`.



Activity 1 — Draw P

P: max $3x_1 + 4x_2$ **s.t.** $x_1 + 2x_2 \leq 8$, $2x_1 + x_2 \leq 9$, $x \in \mathbb{Z}^2_{\geq 0}$

Your turn — The feasible region of P 6 min — GeoGebra Lab 1 ↗ + Miro **1 – Graphical**

1. Type the two constraints of **P** (`x+2y<=8`, ...) and the axes `x>=0`, `y>=0`.
2. Integer points: `Sequence(Sequence((i,j),i,0,5),j,0,5)`.
3. Slider `k` and the line `3x+4y=k`: find the LP optimum.
4. Round the LP optimum. Is the rounded point feasible? Is it the best integer point?
5. **Post on Miro**: a screenshot of your plot, the LP optimum, and the best integer point you see.

GeoGebra: GeoGebra lab ↗

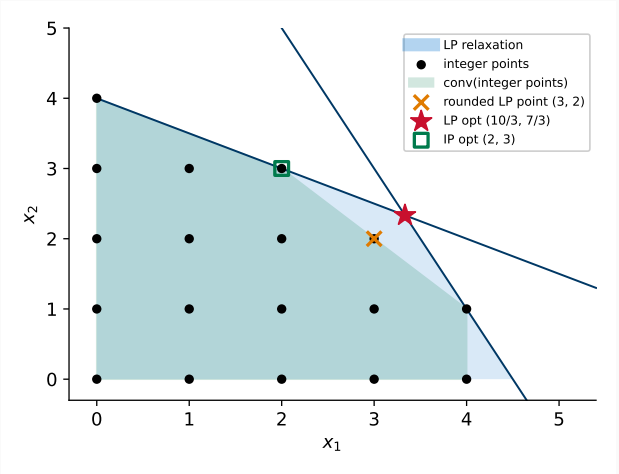
Activity 1 — Solution

P: max $3x_1 + 4x_2$ **s.t.** $x_1 + 2x_2 \le 8,\ 2x_1 + x_2 \le 9,\ x \in \mathbb{Z}^2_{\ge 0}$

- LP optimum: $x^* = (\frac{10}{3}, \frac{7}{3})$, $z^* = \frac{58}{3} \approx 19.33$
- Rounded: $(3, 2)$, feasible, $z = 17$
- Best integer point: $(2, 3)$, $z = 18$

Lesson

Rounding can be feasible *and still not optimal*. We need a method that **proves** $(2, 3)$ is optimal: the bound 19.33 must come down to 18.



2. The simplex method in matrix form

Simplex: a walk on the extreme points

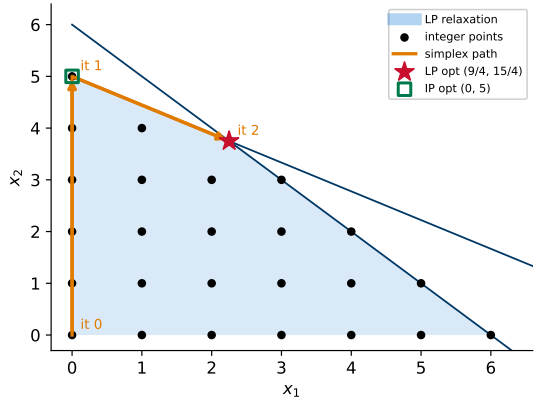
E: max $5x_1 + 8x_2$ **s.t.** $x_1 + x_2 \leq 6,\ 5x_1 + 9x_2 \leq 45,\ x \in \mathbb{Z}^2_{\geq 0}$

- If an LP has an optimum, one of its **extreme points** is optimal.
- Simplex moves from an extreme point to a **better neighbour** (along an edge).
- It stops when no neighbour is better: optimal.

Where do we start?

With $Ax \leq b$ and $b \geq 0$: the **origin** is an extreme point (all slacks basic, $B = I$).

Otherwise ($\geq$ or $=$ rows, $b_i < 0$): **Phase I** (minimize artificial variables) or **big-M** finds a first extreme point.



E: $(0,0) \to (0,5) \to (\frac{9}{4}, \frac{15}{4})$

The simplex method in matrix form

Add slacks: $[A \mid I]\binom{x}{s} = b$. A **basis** $= m$ columns, matrix B (variables x_B); the other columns form H (non-basic variables x_H).

Multiply $Bx_B + Hx_H = b$ by B^{-1}

$$x_B + B^{-1}H\,x_H = B^{-1}b$$

$$z = c_B B^{-1} b + \bar{c}_H\, x_H$$

$$\bar{c}_H = c_H - c_B B^{-1} H \text{ (reduced costs)}$$

One equation per basic variable, plus z written with x_H only. The coefficients of the z row are the *reduced costs*.

Solvers never invert B from scratch: they *update* a factorization of B after each swap.

Read everything from the equations

1. **Point**: $x_H = 0 \Rightarrow x_B = B^{-1}b$, $z =$ the constant
2. **Optimal** (max) if no reduced cost is positive
3. **Enter**: *you choose* any x_j with $\bar{c}_j > 0$ (common rule: the largest)
4. **Leave**: increase it; each row $x_{B(i)} = \bar{b}_i - \bar{a}_{ij}x_j \geq 0$ limits it; the first to hit 0 leaves (ratio test)
5. **New basis**: swap one column of B, new B^{-1}, new equations

The final rows are the raw material of **Gomory cuts** (Part 4). Live demo, notebook 01↗: `E_enter = []`, then `["x2"]`, then `["x2", "x1"]`.

E step by step: iteration 0

E: max $5x_1 + 8x_2$ **s.t.** $x_1 + x_2 \le 6,\ 5x_1 + 9x_2 \le 45,\ x \in \mathbb{Z}^2_{\ge 0}$

Basis

$x_B = (s_1, s_2), \quad x_H = (x_1, x_2), \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 1 \\ 5 & 9 \end{pmatrix}, \quad B^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

Equations $x_B + B^{-1}H\,x_H = B^{-1}b$

$$s_1 + x_1 + x_2 = 6$$
$$s_2 + 5x_1 + 9x_2 = 45$$
$$z = 0 + 5x_1 + 8x_2$$

Reduced costs $\bar{c}_H = c_H - c_B B^{-1} H$

$c_B = (0,0), \quad c_H = (5,8) \quad \Rightarrow \quad \bar{c}_H = (5,8)$

Solution ($x_H = 0$)

$x_B = (s_1, s_2) = (6, 45), \quad x_H = (0,0)$

$\Rightarrow x_1 = 0,\ x_2 = 0,\ s_1 = 6,\ s_2 = 45$

point $(0,0), \quad z = 0$

Next basis

Both $\bar{c} > 0$ can enter; we take the largest: x_2 **enters**.

$s_1 = 6 - x_2 \ge 0 \Rightarrow x_2 \le 6$

$s_2 = 45 - 9x_2 \ge 0 \Rightarrow x_2 \le \mathbf{5}$

s_2 hits 0 first: s_2 **leaves**, x_2 takes its place.

E step by step: iteration 1

E: max $5x_1 + 8x_2$ **s.t.** $x_1 + x_2 \le 6,\ 5x_1 + 9x_2 \le 45,\ x \in \mathbb{Z}^2_{\ge 0}$

Basis

$x_B = (s_1, x_2),\quad x_H = (x_1, s_2),\qquad B = \begin{pmatrix} 1 & 1 \\ 0 & 9 \end{pmatrix},\quad H = \begin{pmatrix} 1 & 0 \\ 5 & 1 \end{pmatrix},\quad B^{-1} = \begin{pmatrix} 1 & -\frac{1}{9} \\ 0 & \frac{1}{9} \end{pmatrix}$

Equations $x_B + B^{-1}H\,x_H = B^{-1}b$

$$s_1 + \tfrac{4}{9}x_1 - \tfrac{1}{9}s_2 = 1$$
$$x_2 + \tfrac{5}{9}x_1 + \tfrac{1}{9}s_2 = 5$$
$$z = 40 + \tfrac{5}{9}x_1 - \tfrac{8}{9}s_2$$

Reduced costs $\bar{c}_H = c_H - c_B B^{-1} H$

$c_B = (0, 8),\quad c_H = (5, 0)\quad \Rightarrow\quad \bar{c}_H = (\tfrac{5}{9}, -\tfrac{8}{9})$

$(0, 5)$ is the IP optimum, but the LP does not care about integrality: it moves on.

Solution ($x_H = 0$)

$x_B = (s_1, x_2) = (1, 5),\quad x_H = (0, 0)$

$\Rightarrow x_1 = 0,\ x_2 = 5,\ s_1 = 1,\ s_2 = 0$

point $(0, 5),\quad z = 40$

Next basis

Only $\bar{c}_{x_1} > 0$: x_1 **enters**.

$s_1 = 1 - \frac{4}{9}x_1 \ge 0 \Rightarrow x_1 \le \frac{9}{4}$

$x_2 = 5 - \frac{5}{9}x_1 \ge 0 \Rightarrow x_1 \le 9$

s_1 hits 0 first: s_1 **leaves**, x_1 takes its place.

E step by step: iteration 2 (optimal)

E: max $5x_1 + 8x_2$ **s.t.** $x_1 + x_2 \le 6,\ 5x_1 + 9x_2 \le 45,\ x \in \mathbb{Z}^2_{\ge 0}$

Basis

$x_B = (x_1, x_2), \quad x_H = (s_1, s_2), \qquad B = \begin{pmatrix} 1 & 1 \\ 5 & 9 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad B^{-1} = \frac{1}{4}\begin{pmatrix} 9 & -1 \\ -5 & 1 \end{pmatrix}$

Equations $x_B + B^{-1}H\,x_H = B^{-1}b$

$$x_1 + \tfrac{9}{4}s_1 - \tfrac{1}{4}s_2 = \tfrac{9}{4}$$
$$x_2 - \tfrac{5}{4}s_1 + \tfrac{1}{4}s_2 = \tfrac{15}{4}$$
$$z = \tfrac{165}{4} - \tfrac{5}{4}s_1 - \tfrac{3}{4}s_2$$

Reduced costs $\bar{c}_H = c_H - c_B B^{-1} H$

$c_B = (5, 8), \quad c_H = (0, 0) \quad \Rightarrow \quad \bar{c}_H = (-\tfrac{5}{4}, -\tfrac{3}{4})$

Keep these rows for Part 4: they are the raw material for **Gomory cuts**.

Solution ($x_H = 0$)

$x_B = (x_1, x_2) = (\tfrac{9}{4}, \tfrac{15}{4}), \quad x_H = (0, 0)$

$\Rightarrow x_1 = \tfrac{9}{4},\ x_2 = \tfrac{15}{4},\ s_1 = 0,\ s_2 = 0$

point $(\tfrac{9}{4}, \tfrac{15}{4}), \quad z = \tfrac{165}{4}$

Next basis: none, optimal

No positive reduced cost: increasing s_1 or s_2 can only decrease z.

Activity 2 — Simplex on P

P: max $3x_1 + 4x_2$ **s.t.** $x_1 + 2x_2 \leq 8$, $2x_1 + x_2 \leq 9$, $x \in \mathbb{Z}^2_{\geq 0}$

Your turn — Follow the simplex path of P 7 min — Colab 01↗ + Lab 2↗ + Miro **2 – Simplex**

1. In notebook 01↗ (**P** is already written in PuLP), build `P_enter` one variable at a time: start with `[]`, read the reduced costs $\bar{c}_H$, add a variable with $\bar{c}_j > 0$, run again… until **optimal**.
2. **Post on Miro**: your list, the extreme points visited, and the optimal basis, B, B^{-1}, x_B, z.
3. In one sentence: **why** does the method stop where it stops?

Concept check: at each iteration, which variables *could* enter, and which one did you choose? Which constraint stops the move (ratio test)? Compare with a neighbour who chose differently.

Activity 2 — Solution

P: max $3x_1 + 4x_2$ **s.t.** $x_1 + 2x_2 \le 8,\ 2x_1 + x_2 \le 9,\ x \in \mathbb{Z}^2_{\ge 0}$

`["x2", "x1"]`:

$$(0,0) \xrightarrow{x_2 \text{ in}, s_1 \text{ out}} (0,4) \xrightarrow{x_1 \text{ in}, s_2 \text{ out}} \left(\tfrac{10}{3}, \tfrac{7}{3}\right)$$

`["x1", "x2"]`:

$$(0,0) \xrightarrow{x_1 \text{ in}, s_2 \text{ out}} \left(\tfrac{9}{2}, 0\right) \xrightarrow{x_2 \text{ in}, s_1 \text{ out}} \left(\tfrac{10}{3}, \tfrac{7}{3}\right)$$

Another route, **same optimum**.

Optimal basis (x_2, x_1):

$$B = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad B^{-1} = \tfrac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

$$x_B = (x_2, x_1) = \left(\tfrac{7}{3}, \tfrac{10}{3}\right), \quad z = \tfrac{58}{3}$$

Order (x_2, x_1): x_2 replaced s_1 in row 1, then x_1 replaced s_2 in row 2.

Final equations: the rows we will cut from

$$\begin{aligned} x_2 + \tfrac{2}{3}s_1 - \tfrac{1}{3}s_2 &= \tfrac{7}{3} \\ x_1 - \tfrac{1}{3}s_1 + \tfrac{2}{3}s_2 &= \tfrac{10}{3} \\ z &= \tfrac{58}{3} - \tfrac{5}{3}s_1 - \tfrac{2}{3}s_2 \end{aligned}$$

Why it stops

Both reduced costs, $\bar{c}_H = (-\tfrac{5}{3}, -\tfrac{2}{3})$, are negative: increasing s_1 or s_2 can only decrease z, so no neighbouring extreme point is better.

3. Branch-and-Bound

Branch-and-Bound: the rationale

Branch

LP solution fractional, $x_j^* = v$: split into $x_j \le \lfloor v \rfloor$ or $x_j \ge \lfloor v \rfloor + 1$.
No integer point is lost; x^* is removed from both children.

Bound

LP value of a node $\ge$ every integer point below it (max).

Prune a node when

- LP is **infeasible**
- LP solution is **integer** (update the incumbent)
- LP bound $\le$ incumbent (**bound**)

Algorithm 1: Branch-and-Bound (max)

$\mathcal{L} \leftarrow \{\text{root}\}$, $\underline{z} \leftarrow -\infty$;
while $\mathcal{L} \neq \emptyset$ **do**
  pick node k; solve its LP $\rightarrow (x^k, z^k)$;
  if *infeasible* **or** $z^k \le \underline{z}$ **then**
    prune; **continue**
  if x^k *integer* **then**
    $\underline{z} \leftarrow z^k$; incumbent $\leftarrow x^k$; **continue**
  choose fractional x_j^k; add two children to $\mathcal{L}$;

Incumbent = lower bound; largest open LP value = upper bound. Stop when they meet.

B&B with PuLP (live demo)

E: max $5x_1 + 8x_2$ **s.t.** $x_1 + x_2 \leq 6,\ 5x_1 + 9x_2 \leq 45,\ x \in \mathbb{Z}^2_{\geq 0}$

Notebook 02 ↗: each node is the LP relaxation with extra bounds on the variables.

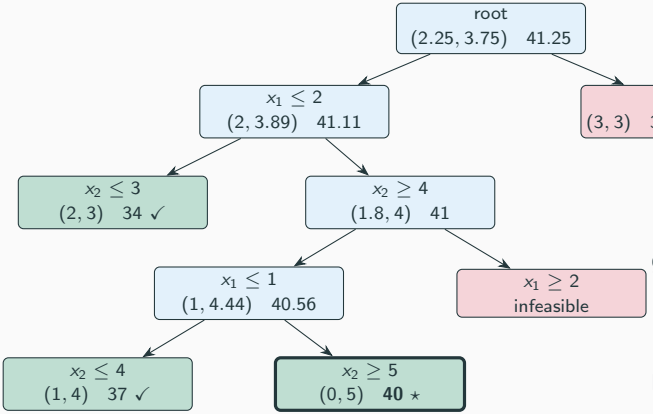
```
E = make_E(integer=True)              # the model, written once in PuLP
E_tree = [                            # (node id, parent id, branching constraint, status)
    (0, None, "root",    "branch"),   # (9/4, 15/4), z = 165/4
    (1, 0,    "x1 <= 2", "?"),        # new node: run, read its LP, decide its status
    (2, 0,    "x1 >= 3", "?"),
]
plot_tree(E, E_tree)                  # solves each node's LP, draws the tree, checks the statuses
```

Each node = the LP relaxation with the bounds of its path; PuLP (HiGHS) solves it. We add the nodes one at a time: exactly what you will do on **P** in Activity 3.

Rule: branch on the first fractional variable, explore the left child ($\leq$) first (depth-first). Status `"?"` = not decided yet.

E: the Branch-and-Bound tree

E: max $5x_1 + 8x_2$ **s.t.** $x_1 + x_2 \leq 6$, $5x_1 + 9x_2 \leq 45$, $x \in \mathbb{Z}^2_{\geq 0}$



9 nodes, 4 LP branchings.

Incumbents: 34 → 37 → 40.

Right child explored last:
pruned by bound (39 $\leq$ 40).

Teaser: with one extra inequality at the root, the tree has 1 node. Which inequality? → Part 4.

Activity 3 — The tree of P

P: max $3x_1 + 4x_2$ **s.t.** $x_1 + 2x_2 \leq 8$, $2x_1 + x_2 \leq 9$, $x \in \mathbb{Z}^2_{\geq 0}$

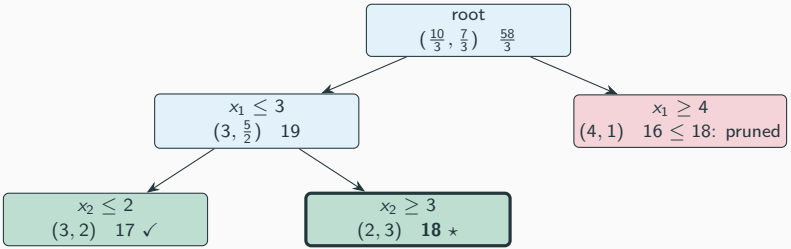
Your turn — Grow the tree of P 8 min — Colab 02↗ + Lab 3↗ + Miro **3 – B&B tree**

1. **Before running anything:** the root LP is $(\frac{10}{3}, \frac{7}{3})$. On which variable do you branch? What are the two children?
2. **Build your tree node by node** in the list `my_tree` (format below) and run `plot_tree(P, my_tree)`: it solves each node's LP, draws the tree and checks your statuses. Add a node with status `"?"`, look at its LP, decide, add its children, re-run.
3. **Post on Miro**: a screenshot of your final tree, and **why** each leaf was closed.
4. **Only then**, compare with the automatic B&B: uncomment the `branch_and_bound(P)` cell.

```
my_tree = [(0, None, "root",    "branch"),   # (node id, parent id, branching constraint, status)
(1, 0,    "x1 <= ?", "?"), ...]  # status:  "?", "branch", "integer", "infeasible", "pruned"
plot_tree(P, my_tree)
```

Activity 3 — Solution

P: max $3x_1 + 4x_2$ **s.t.** $x_1 + 2x_2 \le 8,\ 2x_1 + x_2 \le 9,\ x \in \mathbb{Z}^2_{\ge 0}$



- **5 nodes**. First incumbent $(3, 2)$, $z = 17$: the rounded point.
- $(2, 3)$, $z = 18$ is better: new incumbent.
- $x_1 \ge 4$: integer, but $16 \le 18$: pruned by bound.
- Explore the right child first? Try `left_first=False`.

4. Gomory cuts

Valid inequalities and cuts

Let $P = \{x \geq 0 : Ax \leq b\}$ and $X = P \cap \mathbb{Z}^n$.

Definitions

- $\pi x \leq \pi_0$ is **valid** for X if it holds for every $x \in X$.
- It is a **cut** for x^* if, in addition, $\pi x^* > \pi_0$.
- The best possible: **facets** of $\text{conv}(X)$.

Cutting-plane algorithm

1. Solve LP $\rightarrow x^*$. If integer: stop.
2. **Separation:** find a valid cut violated by x^*.
3. Add it, go to 1.

Branch-and-Cut = B&B + cutting planes

Run the cutting-plane loop *at the nodes* of the tree: tighter bounds $\Rightarrow$ more pruning $\Rightarrow$ smaller trees.

The question of the day

Where do good cuts come from?

- the optimal basis (Gomory)
- rounding combinations (Chvátal–Gomory)
- problem structure (covers, subtours)

Gomory's idea: round a row of the optimal basis

Row i of the final equations $x_B + B^{-1}H\,x_H = B^{-1}b$, with $\bar{a}_{ij} = (B^{-1}H)_{ij}$ and $\bar{b}_i = (B^{-1}b)_i$ fractional: $x_{B(i)} + \sum_{j\in H} \bar{a}_{ij}\,x_j = \bar{b}_i$.

1. Since $x \geq 0$: $\quad x_{B(i)} + \sum_j \lfloor \bar{a}_{ij} \rfloor\, x_j \;\leq\; \bar{b}_i$.
2. The left-hand side is **integer** for integer x, so: $\quad x_{B(i)} + \sum_j \lfloor \bar{a}_{ij} \rfloor\, x_j \;\leq\; \lfloor \bar{b}_i \rfloor$.
3. Subtract from the row; with $f_j = \bar{a}_{ij} - \lfloor \bar{a}_{ij} \rfloor$ and $f_0 = \bar{b}_i - \lfloor \bar{b}_i \rfloor$:

Gomory fractional cut

$$\sum_{j\in H} f_j\,x_j \;\geq\; f_0 \qquad (0 \leq f_j < 1,\; 0 < f_0 < 1)$$

Cuts off x^*: there all non-basic $x_j = 0$, so the left side is $0 < f_0$.

Requires all variables (incl. slacks) integer: true here because A, b are integer. Gomory (1958): with careful rules, the pure cutting-plane method **terminates finitely**.

E: the first cut, from the x_1 row

Row: $x_1 + \frac{9}{4}s_1 - \frac{1}{4}s_2 = \frac{9}{4}$

Fractional parts:
$\text{frac}\left(\frac{9}{4}\right) = \frac{1}{4}, \quad \text{frac}\left(-\frac{1}{4}\right) = \frac{3}{4}, \quad f_0 = \frac{1}{4}$

Cut: $\frac{1}{4}s_1 + \frac{3}{4}s_2 \ge \frac{1}{4} \iff s_1 + 3s_2 \ge 1$

Substitute $s_1 = 6 - x_1 - x_2, \quad s_2 = 45 - 5x_1 - 9x_2$:

G1

$$4x_1 + 7x_2 \le 35$$

Check: at x^*, $4(2.25) + 7(3.75) = 35.25 > 35$ ✓

Careful with negatives

$\text{frac}(a) = a - \lfloor a \rfloor$ is always in $[0, 1)$:
$\text{frac}\left(-\frac{1}{4}\right) = -\frac{1}{4} - (-1) = \frac{3}{4}$, **not** $\frac{1}{4}$.

After adding G1

New LP optimum: $(\frac{7}{3}, \frac{11}{3})$, $z = 41$.
Still fractional $\Rightarrow$ new basis, new rows, new cut.

E: Gomory cuts, row by row (live demo)

E: max $5x_1 + 8x_2$ **s.t.** $x_1 + x_2 \le 6,\ 5x_1 + 9x_2 \le 45,\ x \in \mathbb{Z}^2_{\ge 0}$

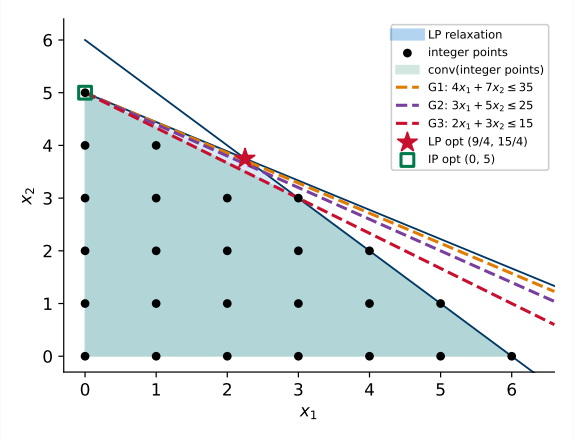
Notebook 03 ↗: we choose the rows one at a time, as for the B&B tree.

```
E_rows = ["x1", "s2", "s3"]
plot_cuts(E, E_rows)
```

Each run: solve the LP after the cuts so far, print its equations and f_0, draw the cuts.

LP point	cut (row)	z
$(\frac{9}{4}, \frac{15}{4})$	G1: $4x_1 + 7x_2 \le 35$ (x_1)	41.25
$(\frac{7}{3}, \frac{11}{3})$	G2: $3x_1 + 5x_2 \le 25$ (s_2)	41
$(\frac{5}{2}, \frac{7}{2})$	G3: $2x_1 + 3x_2 \le 15$ (s_3)	40.5
$(0, 5)$	integer: optimal	40

`E_rows = ["x2"]` gives G3 directly: **the row you pick matters**. G3 passes through $(0, 5)$ and $(3, 3)$: a **facet** of conv(X).



A note on numerical stability

Solvers compute B^{-1} in floating point:

- $x_2 = 3$ comes back as `2.9999999999996`
- $\lfloor \cdot \rfloor$ gives 2, so $f_0 \approx 0.9999999999996$
- an integer row looks fractional, and its "cut" removes the current vertex, an integer point, maybe the optimum
- wrong f_j on the coefficients give invalid cuts

Also: `49 * (1/49) = 0.9999999999999999`. Try it in notebook 03↗.

What solvers do

- tolerances: $\text{frac}(a) < \varepsilon$ or $> 1 - \varepsilon$ counts as integer ($\varepsilon \approx 10^{-6}$)
- relax the right-hand side slightly (safety margin)
- reject **dense** cuts and cuts with large **dynamism** ($\max |a_j| / \min |a_j|$)
- few rounds, mostly at the root; mixed-integer version (GMI)
- numerically safe cuts: Cook, Dash, Fukasawa & Goycoolea (2009)

Our notebook uses exact fractions: fine for 2 variables, not for 10,000.

Activity 4 — Gomory on P

P: max $3x_1 + 4x_2$ **s.t.** $x_1 + 2x_2 \leq 8$, $2x_1 + x_2 \leq 9$, $x \in \mathbb{Z}^2_{\geq 0}$

Your turn — Cut P until the LP is integer 7 min — Colab 03↗ + Lab 4↗ + Miro **4 – Gomory**

1. `basis_report(P)`: do you recognize B^{-1} from Activity 2?
2. **Build your list of rows** `P_rows`, one row at a time: start with `[]`, read the equations, pick a fractional row, add it, re-run `plot_cuts(P, P_rows)` until the LP point is integer.
3. **Post on Miro**: a **screenshot of the graph with the cuts**, and on a sticky **your answers**:
   - your list of rows, the cuts and the final LP solution;
   - does the **first** cut pass through an integer point? What do you notice about its coefficients?
   - B&B nodes (Activity 3) vs. number of cuts.

Activity 4 — Solution

P: max $3x_1 + 4x_2$ **s.t.** $x_1 + 2x_2 \le 8,\ 2x_1 + x_2 \le 9,\ x \in \mathbb{Z}^2_{\ge 0}$

Row $x_2 + \frac{2}{3}s_1 - \frac{1}{3}s_2 = \frac{7}{3}$:
$f = (\frac{2}{3}, \frac{2}{3})$, $f_0 = \frac{1}{3} \Rightarrow 2s_1 + 2s_2 \ge 1$

G1: $2x_1 + 2x_2 \le 11 \to$ LP $(3, \frac{5}{2})$, $z = 19$

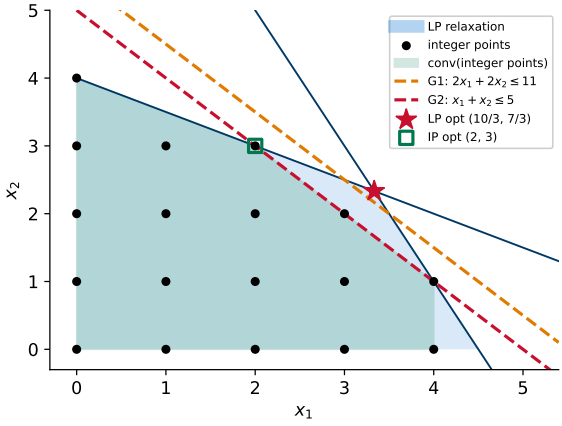
G2: $x_1 + x_2 \le 5 \quad \to$ LP $(2, 3)$, $z = 18$ ✓

Observe

G1 touches **no** integer point: its left side is even, so ≤ 11 means ≤ 10. Divide by 2 and round: $x_1 + x_2 \le 5.5 \to \le 5 =$ G2, a facet.

Rounding after combining: **Part 5**.

Any order of rows (e.g. `["x2", "s2"]`) gives these two cuts. B&B: 5 nodes. Gomory: 2 cuts, then one LP.



Gomory cuts in practice

Why they were abandoned (1960s–80s)

- Pure cutting-plane loop: slow tail-off
- Dense cuts, numerical trouble
- Each cut derived from the previous basis

Why they came back (Balas et al., 1996)

- Add *rounds* of cuts at the root only, then branch
- Gomory **mixed-integer** (GMI) version for MIPs
- Today: in every solver (CPLEX, Gurobi, HiGHS, CBC, SCIP)

Cutting *and* branching together is Branch-and-Cut: a few cuts shrink the tree; branching guarantees the end.

5. Chvátal–Gomory cuts

The Chvátal–Gomory (CG) procedure

No basis needed: combine the constraints, then round.

CG cut: for any multipliers $u \geq 0$

$$\underbrace{u^\top A x \leq u^\top b}_{\text{valid for } P} \;\Rightarrow\; \underbrace{\lfloor u^\top A \rfloor\, x \leq u^\top b}_{x \geq 0} \;\Rightarrow\; \boxed{\lfloor u^\top A \rfloor\, x \leq \lfloor u^\top b \rfloor} \quad \text{valid for } X = P \cap \mathbb{Z}^n$$

E with $u = (\frac{1}{2}, \frac{1}{2})$

$\frac{1}{2}(x_1+x_2) + \frac{1}{2}(5x_1+9x_2) \leq \frac{1}{2}(6+45)$

$3x_1 + 5x_2 \leq 25.5 \;\Rightarrow\; 3x_1 + 5x_2 \leq 25 \;\; (= \text{G2!})$

At x^*: $6.75 + 18.75 = 25.5 > 25$ ✓

Key facts

- Every **Gomory cut is a CG cut**: take $u = \text{frac}(\text{row of } B^{-1})$.
- **Chvátal closure** P': P + all CG cuts. Repeat: $P \supseteq P' \supseteq P'' \supseteq \ldots$
- **Chvátal (1973):** for a rational polytope, $P^{(t)} = \text{conv}(X)$ for some finite t (the *CG rank*).
- Separating CG cuts in general is NP-hard ⇒ solvers use special multipliers.

Activity 5 — Multiplier hunt

Your turn — Find the multipliers 6 min — paper, Colab 03 ↗ + Lab 5 ↗ + Miro **5 – CG hunt**

E: $x_1 + x_2 \le 6$ (u_1), $5x_1 + 9x_2 \le 45$ (u_2)

1. Find $u \ge 0$ giving $2x_1 + 3x_2 \le 15$, and u giving $4x_1 + 7x_2 \le 35$. *Hint: solve $u^\top A =$ target coefficients, then round the RHS.*
2. Compare your u's with the rows of $B^{-1} = \frac{1}{4}\begin{pmatrix} 9 & -1 \\ -5 & 1 \end{pmatrix}$. What do you notice?
3. **P**: find u giving $x_1 + x_2 \le 5$ in **one** step.
4. **E**: $u = (0, \frac{1}{9})$ gives a valid inequality. Is it a *cut* for $x^* = (\frac{9}{4}, \frac{15}{4})$?

Check each answer with `cg_cut(model, u, x_star)` in notebook 03 ↗; post your u's on Miro.

Activity 5 — Solution

1. $u_1 + 5u_2 = 2,\ u_1 + 9u_2 = 3 \Rightarrow u = (\frac{3}{4}, \frac{1}{4})$, RHS $15.75 \to 15$.
   $u = (\frac{1}{4}, \frac{3}{4})$: RHS $= 35.25 \to 35$.
2. $\text{frac}\big((\frac{9}{4}, -\frac{1}{4})\big) = (\frac{1}{4}, \frac{3}{4})$ and $\text{frac}\big((-\frac{5}{4}, \frac{1}{4})\big) = (\frac{3}{4}, \frac{1}{4})$:
   exactly the fractional parts of the rows of B^{-1}!
3. $u = (\frac{1}{3}, \frac{1}{3})$: $x_1 + x_2 \le \frac{17}{3} \to 5$. One CG round reaches $\text{conv}(X)$ of **P**.
4. $u = (0, \frac{1}{9})$: $\lfloor (\frac{5}{9}, 1) \rfloor = (0, 1)$, RHS $= 5$: $x_2 \le 5$.
   Valid, but $x_2^* = 3.75 \le 5$: not a cut for x^*.

Take-away

Gomory = CG with multipliers *read off the basis*.

CG = rounding *any* nonnegative combination: more freedom, but finding a violated one is hard.

Rank

For **E** and **P** a single round of CG cuts gives $\text{conv}(X)$: rank 1. Some polytopes need very high rank.

Break — 5 minutes

Next: knapsack covers and the TSP.

Open notebook `05_TSP_Branch_and_Cut.ipynb` ↗ in Colab while you wait.

6. Knapsack & cover inequalities

Exploiting structure: cover inequalities

0–1 knapsack

$$\begin{aligned} \max\ & 8x_1 + 11x_2 + 6x_3 + 4x_4 \\ \text{s.t. } & 5x_1 + 7x_2 + 4x_3 + 3x_4 \le 14 \\ & x \in \{0,1\}^4 \end{aligned}$$

LP: $x^* = (1, 1, \frac{1}{2}, 0)$, $z^* = 22$. Optimum: $(0, 1, 1, 1)$, $z = 21$.

Cover $\{1, 2, 3\}$ ($5{+}7{+}4 = 16 > 14$): $x_1 + x_2 + x_3 \le 2$, violated ($2.5 > 2$). New LP: $z = 21.\overline{6}$; profits are integers $\Rightarrow z \le 21$: **optimal, no branching**.

Cover inequalities

C is a **cover** if $\sum_{j \in C} a_j > b$: not all items of C fit, so

$$\sum_{j \in C} x_j \le |C| - 1.$$

Separation = another knapsack

$\zeta = \min\{\sum_j (1 - x_j^*) z_j : \sum_j a_j z_j \ge b + 1,\ z \in \{0,1\}^n\}$;
$\zeta < 1 \iff$ the cover $\{j : z_j = 1\}$ is violated.

Every row of a 0–1 program is a knapsack: covers (plus *lifting*) are separated row by row in every solver (Crowder, Johnson & Padberg, 1983). Exercises: notebook 04 ↗ (at home).

Strengthening covers: lifting

A cover inequality gives coefficient 0 to every item outside C: the cut is weak.

Sequential up-lifting

Start from $\sum_{i \in C} x_i \le |C| - 1$. For each $j \notin C$, in turn, give x_j the largest coefficient that keeps the inequality valid:

$$\alpha_j = |C| - 1 - m_j, \quad \text{with}$$
$$m_j = \max_{x \in \{0,1\}} \left\{ \textstyle\sum_{i \in L} \alpha_i x_i : \sum_{i \in L} a_i x_i \le b - a_j \right\}$$

L = items **already lifted** (the cover, then each x_j once its α_j is computed): their α_i are known numbers, so m_j is a **linear** 0–1 knapsack in x. One item at a time; the order can change the result.

- Minimal cover + sequential lifting $\Rightarrow$ a **facet** of conv(X) (Balas, 1975).
- Made 0–1 programming practical (Crowder, Johnson & Padberg, 1983). At home: notebook 04 ↗.

Example (Nemhauser & Wolsey)

$11x_1 + 6x_2 + 6x_3 + 5x_4 + 5x_5 + 4x_6 + x_7 \le 19$

$C = \{3, 4, 5, 6\}$: $6+5+5+4 = 20 > 19$ (minimal) $\Rightarrow$ $x_3 + x_4 + x_5 + x_6 \le 3$

lift	L (with known α)	cap. $19 - a_j$	m_j	$\alpha_j = 3 - m_j$
x_1	$\{3, 4, 5, 6\}$, all 1	8	1	**2**
x_2	$+ x_1$ with $\alpha_1 = 2$	13	2	**1**
x_7	$+ x_2$ with $\alpha_2 = 1$	18	3	**0**

Lifted cover: $\boxed{2x_1 + x_2 + x_3 + x_4 + x_5 + x_6 \le 3}$

Why it matters

$x^* = (1, 0, 1, 0, 0, \frac{1}{2}, 0)$, weight 19: in the LP relaxation.
Cover: $1 + \frac{1}{2} \le 3$, **not** cut off. Lifted: $2 + 1 + \frac{1}{2} = 3.5 > 3$, **cut off**.

7. Case study: the Traveling Salesperson Problem

The classic model: directed TSP + MTZ

Arc variable $x_{ij} = 1$ if the tour goes **from** i **to** j ($i \neq j$); city 1 is the start.

Miller–Tucker–Zemlin (1960)

$$\min \sum_{i \neq j} c_{ij} x_{ij}$$

$$\text{s.t.} \sum_{j \neq i} x_{ij} = 1, \quad \sum_{j \neq i} x_{ji} = 1 \quad \forall i \quad \text{(leave, enter)}$$

$$u_i - u_j + n\, x_{ij} \leq n - 1 \qquad \forall i \neq j,\ i, j \geq 2$$

$$2 \leq u_i \leq n, \quad x_{ij} \in \{0, 1\}$$

u_i = **position** of city i in the tour. If arc (i, j) is used, the MTZ constraint gives $u_j \geq u_i + 1$; otherwise it is inactive ($u_i - u_j \leq n - 2$ always holds). A subtour avoiding city 1 would need u to increase all around a cycle: **impossible**.

Good

Compact: $n(n-1)$ binary $+ (n-1)$ continuous variables, $\approx n^2$ constraints. Any MIP solver takes it as it is.

Bad

The $n\, x_{ij}$ term is a **big-M**: the LP relaxation is very weak (fractional x make the MTZ constraints almost free). Large gaps and large B&B trees: it struggles quickly as n grows.

Next: drop the direction and the u variables, and pay with exponentially many constraints.

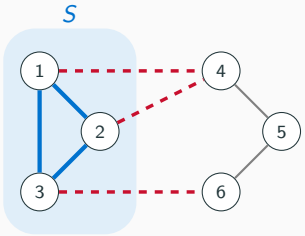
TSP notation

- $G = (V, E)$: complete undirected graph, $V = \{1, \ldots, n\}$ cities, $|E| = \frac{n(n-1)}{2}$
- Edge $e = \{i, j\}$ with length $c_e = c_{ij}$
- Variable $x_e \in \{0, 1\}$: 1 if the tour uses edge e
- For an edge set F: $\quad x(F) = \sum_{e \in F} x_e$
- $\delta(i)$: edges **incident** to city i
- $\delta(S)$: edges with **exactly one** end in S (the *cut* of S)
- $E(S)$: edges with **both** ends in S

Two ways to write a SEC

Sum the degree equations over $i \in S$:
$\sum_{i \in S} x(\delta(i)) = 2\,x(E(S)) + x(\delta(S)) = 2|S|$, so

$$x(\delta(S)) \geq 2 \iff x(E(S)) \leq |S| - 1.$$



solid blue: $E(S)$
dashed red: $\delta(S)$

Reading a subtour

Suppose the solution is the two triangles $1-2-3$ and $4-5-6$ (degrees all $= 2$). For $S = \{1, 2, 3\}$: $x(E(S)) = 3 > 2$ and $x(\delta(S)) = 0 < 2$, so the SEC is violated.

The symmetric TSP: DFJ formulation

Notation of the previous slide: $x_e = 1$ if edge e is in the tour.

Dantzig–Fulkerson–Johnson (1954)

$$\begin{aligned} \min\ & \textstyle\sum_{e \in E} c_e x_e \\ \text{s.t. } & x(\delta(i)) = 2 && \forall i \in V \\ & x(\delta(S)) \geq 2 && \forall S \subset V,\ 2 \leq |S| \leq n-2 \\ & x_e \in \{0,1\} && \forall e \in E \end{aligned}$$

Degree: enter and leave every city exactly once. Alone, this allows several disjoint cycles.
SEC (subtour elimination): every proper group of cities is left by ≥ 2 tour edges. $|S| = 1$ is already covered by the degree equation, and S and $V \setminus S$ give the same cut.

Why not MTZ here?

An edge $\{i, j\}$ has **no direction**: "$u_j \geq u_i + 1$ if the tour goes from i to j" has no meaning. Subtours must be cut by SECs.

	directed + MTZ	symmetric DFJ
variables	$n(n-1) + (n-1)$	$n(n-1)/2$
constraints	$\approx n^2$	$\approx 2^n$
LP bound	weak	strong (1–2% gap)

The catch

2^n SECs: $n = 50$ gives $\approx 10^{15}$. We can't write them down.

The trick

Start *without* SECs and add only the ones violated by the current solution: **separation** again!

Activity 6a — Be the separation oracle

Your turn — Cut the subtours yourself (16 cities) 7 min — Colab 05↗ + Miro frame **6 – TSP results**

1. **Your own instance**: set `MY_SEED` (not 5, used on the slides) and run `new_instance`: it draws your 16 cities.
2. In the **solve cell**, `SETS = []` and `solve_with_cuts(SETS)`: **degree constraints only**.
3. Not a single tour? Add **one** subtour to `SETS`, e.g. `[[0, 7, 12]]`, then `[[0, 7, 12], [4, 6, 13]]`, ..., and **re-run the same cell**: the cut $x(\delta(S)) \geq 2$ is added, the model re-solved and re-drawn. **One subtour per round**, until you get one tour.
4. **Post on Miro**: your final picture (it shows your seed and your sets) and the printed `SETS = ...` line.

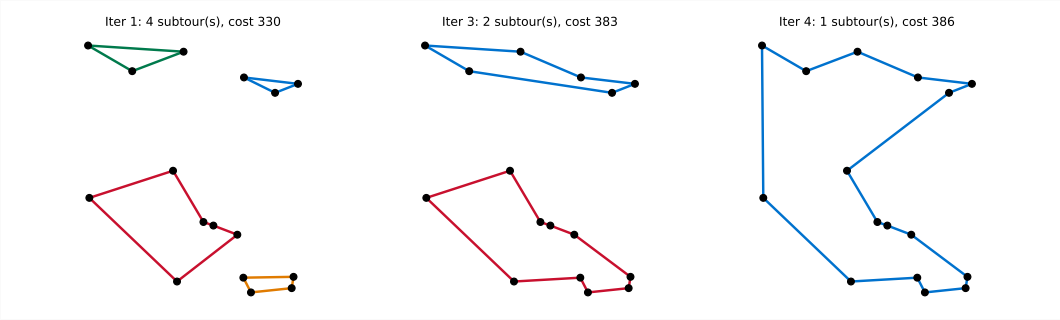
Count your rounds

How many rounds (solves) did you need? Next slide: one subtour per round vs. all at once.

Think about it

How did you spot a subtour in the picture? Write your procedure in words on Miro.

Activity 6a — One run, two strategies



Strategy	MIP solves	SECs added	optimal cost
cut **all** subtours each round	4	8	385.7
cut **one** (smallest) per round	6	5	385.7

The cost never decreases ($330 \to \cdots \to 386$): each round only *adds* constraints, so every round is a valid **lower bound**. Fewer cuts per round means more re-solves; more cuts means a bigger model.

From your eyes to an oracle

What you did with your eyes: follow the edges from a city until you come back, then check whether everyone was visited.

Integer x^*: graph search (lazy constraints)

$G^* = (V, \{e : x_e^* = 1\})$ is a union of disjoint cycles.
Subtour = **connected component** $S \neq V$ of G^*; then $x^*(\delta(S)) = 0$.
Find all of them with **DFS/BFS** in $O(n + |E^*|) = O(n)$.
Tour $\iff$ G^* has **one** component.

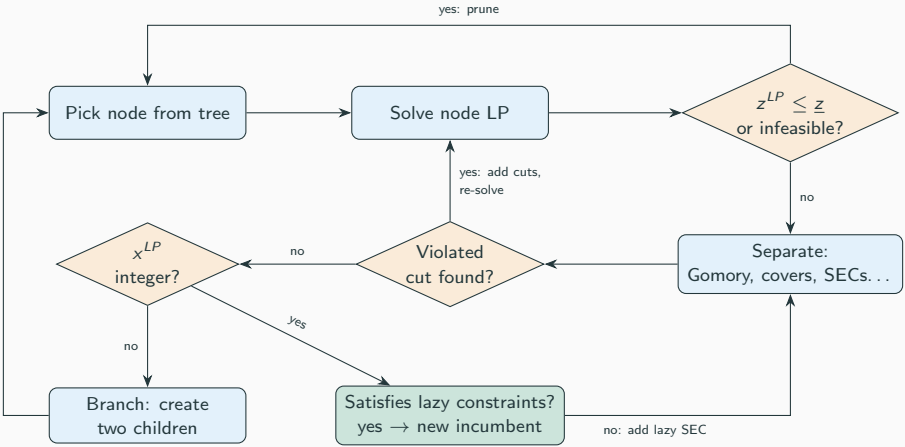
Fractional x^*: min cut (user cuts)

The support graph can be *connected* and still have $x^*(\delta(S)) < 2$.
Give edge e capacity x_e^* and compute a **global minimum cut** $\delta(S)$:
value < 2 $\iff$ violated SEC.
Cost: polynomial (Stoer–Wagner, or $n-1$ max-flows).

Separation $\equiv$ optimization (Grötschel–Lovász–Schrijver)

SECs can be separated in polynomial time, so the LP over *all* 2^n SECs is solvable in polynomial time: exponentially many constraints are fine with a good oracle.

The full Branch-and-Cut loop



User cuts tighten the LP (optional, for speed). **Lazy constraints** are part of the model (needed for correctness, e.g. SECs on integer solutions).

Branch-and-Cut in a solver: lazy constraints

With Gurobi, SECs are added *inside* one B&B search instead of re-solving from scratch:

```
def subtour_cb(model, where):
    if where == GRB.Callback.MIPSOL:                    # integer solution found
        vals = model.cbGetSolution(model._x)
        G = nx.Graph([e for e in model._x if vals[e] > 0.5])
        for S in nx.connected_components(G):
            if len(S) < model._n:                       # subtour -> lazy SEC
                model.cbLazy(gp.quicksum(model._x[i, j] for i, j in model._x
                                         if i in S and j in S) <= len(S) - 1)

m.Params.LazyConstraints = 1
m.optimize(subtour_cb)
```

Notebook 05↗ runs the same idea with free tools (PuLP + HiGHS), and has an optional Gurobi cell (free academic license).

Activity 6b — Code your oracle

Your turn — Automate what you did by hand 8 min — Colab 05 ↗ + Miro frame **6 – TSP results**

1. Write `find_subtours(n, chosen)` **yourself** (no networkx): DFS/BFS from every unvisited city. Test it on the two given examples.
2. Plug it into `tsp_integer_loop`: same cost as in 6a? Run $n = 20, 40, 60$ and post rounds, SECs, cost and time on Miro.
3. **If time:** fractional x^*. Write the min-cut oracle with the AI prompt below, then compare the degree-only LP, the subtour LP and the optimum.

```
"Using networkx, write separate_sec_fractional(n, x_star) where x_star maps edges (i, j), i<j, of a
complete graph on n nodes to LP values.  Build the support graph with capacity x_star[e] for x_star[e]
> 1e-6, and if it is disconnected return each component as a violated set S. Otherwise compute a global
min cut with nx.stoer_wagner (weight='capacity'); if its value < 2 - 1e-6 return [S] for the smaller
side, else return [].  Explain why x(delta(S)) < 2 means a violated subtour elimination constraint."
```

Activity 6b — What you should observe

Integer loop

- Few rounds, but each round is a full MIP solve: time grows quickly with n.
- Real B&C (lazy callbacks) does it in *one* tree.

LP bounds

- Degree-only LP: noticeably weaker.
- Degree + all SECs (*subtour bound*): typically within 1–2% of the optimum.

Beyond SECs

Concorde (Applegate, Bixby, Chvátal, Cook) adds **comb**, blossom and other facet classes, and solved an 85,900-city instance to optimality.

Same recipe everywhere

model → relax → separate → branch.
Vehicle routing, network design, scheduling...

Wrap-up

Take-aways

1. Branch-and-Cut is an **exact** method: it returns a solution *and* a proof (gap $= 0$).
2. The LP relaxation gives a **bound**; the simplex method finds it by walking along extreme points, and its optimal basis B^{-1} is the raw material for cuts.
3. B&B closes the gap by **enumeration**; cuts close it by **tightening**. Together: Branch-and-Cut.
4. **Gomory** cuts come from rows of B^{-1}; they are **Chvátal–Gomory** cuts. Mind the **floating point**.
5. **Structure wins**: covers for knapsacks, SECs for the TSP; exponentially many constraints are fine if you can **separate** them.

Exit ticket (Miro, 3 min) & resources

Your turn — Exit ticket 3 min — Miro **7 – Exit**

1. One thing you can now do that you couldn't 2 hours ago.
2. One thing still unclear.
3. The notebooks + live demos were: *helpful / neutral / confusing*.

To go further

- L. Wolsey, *Integer Programming*, 2nd ed., Wiley, 2020.
- M. Conforti, G. Cornuéjols, G. Zambelli, *Integer Programming*, Springer, 2014.
- D. Applegate et al., *The Traveling Salesman Problem: A Computational Study*, Princeton, 2006.
- W. Cook, *In Pursuit of the Traveling Salesman*, Princeton, 2012.

Thank you!

Angélica Salazar (UANL)
Vincent Boyer (Tecnológico de Monterrey)

Questions: maria.salazaragl@uanl.edu.mx · vincent.boyer@tec.mx

Backup — Gomory mixed-integer (GMI) cut

For a row $x_{B(i)} + \sum_{j \in H} \bar{a}_j x_j = \bar{b}$ where some non-basic x_j are continuous, with $f_0 = \mathrm{frac}(\bar{b})$ and $f_j = \mathrm{frac}(\bar{a}_j)$:

$$\sum_{\substack{j \in I \\ f_j \le f_0}} \frac{f_j}{f_0} x_j + \sum_{\substack{j \in I \\ f_j > f_0}} \frac{1 - f_j}{1 - f_0} x_j + \sum_{\substack{j \in C \\ \bar{a}_j > 0}} \frac{\bar{a}_j}{f_0} x_j - \sum_{\substack{j \in C \\ \bar{a}_j < 0}} \frac{\bar{a}_j}{1 - f_0} x_j \;\ge\; 1$$

I = integer non-basic, C = continuous non-basic. It dominates the fractional cut on pure IPs and is the workhorse Gomory cut inside commercial solvers.

Backup — Home exercises

1. On **E**, start the Gomory loop from the x_2 row (`gomory_cut(E, row="x2")`). How many cuts?
2. Write your own 3-variable IP in PuLP and run `gomory_loop` (notebook 03 ↗). How large do the coefficients get?
3. Notebook 04 ↗: separate knapsack covers by hand, then lift them.
4. Show that for the TSP, $x(\delta(S)) \geq 2$ and $x(E(S)) \leq |S| - 1$ are equivalent given the degree equations.
5. Run notebook 05 ↗ (Gurobi callback or HiGHS loop) for $n = 60$ and compare with the MTZ formulation.